Task 01: Evaluate Requirements of Each Computing Environment

Operating Systems (CSA04) — Assignment 1 (CO1)

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Marks: 4 Marks (Bloom's: K5 - Evaluate)
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> ***Task 1 Description:*** *Evaluate what each part of the hospital needs from its operating system. Discuss response-time, reliability, throughput, battery, and security requirements; analyze conflicting requirements and patient safety impacts; examine lessons from OS history; rank requirements; and analyze the alternative of a single general-purpose OS.*

1. RANKED REQUIREMENT MATRIX

Environment	Ranked Priority	Key Metrics & Constraints	Clinical & System Justification
Bedside Monitors	1. Response Time 2. Reliability 3. Security 4. Throughput 5. Battery	Hard Real-Time: Alarm trigger ≤ 10ms. Availability: 99.999% (five-nines). Zero unhandled faults.	Direct patient safety. Missing an arrhythmia or asystole threshold by 50ms results in irreversible cardiac or cerebral damage.
Nurses' Tablets	1. Battery Life 2. Reliability 3. Security 4. Response Time 5. Throughput	Battery: ≥ 12-hour full nursing shift. Latency: < 200ms UI response. AES-256 storage sandbox.	Mid-shift device shutdown delays urgent medication rounds. High mobile theft/loss risk demands encrypted local cache.
Doctors' Workstations	1. Security 2. Interactivity 3. Reliability 4. Throughput 5. Battery	UI latency: < 100ms. Multi-role authentication (Smart Card / FIDO2). POSIX multi-window stack.	HIPAA/audit compliance is primary. Doctors rapidly alternate between EHR charting, diagnostic PACS viewing, and tele-health.
Central Database	1. Reliability 2. Throughput 3. Security 4. Response Time 5. Battery	Availability: 100% 24/7/365 uptime. Throughput: > 5,000 TPS. RPO = 0, RTO < 5s.	Downtime freezes admissions, pharmacy orders, surgical prep, and clinical alerts across all 500 beds simultaneously.
Imaging Server	1. Throughput 2. Memory Space 3. Reliability 4. Response Time 5. Battery	Multi-core parallel throughput: 10–50 GB raw MRI matrix reconstruction in < 60 seconds.	High-bandwidth pipeline utilization avoids backlogs in trauma surgery. Real-time deterministic latency is not required.
Partner Link	1. Security 2. Reliability 3. Interoperability 4. Throughput 5. Response Time	Zero-Trust TLS 1.3 / mTLS mutual auth, HL7/FHIR message validation, defensive rate limiting.	Cross-boundary traffic traverses public internet, exposing data to man-in-the-middle attacks and regulatory violations.

2. CONFLICTING REQUIREMENTS & PATIENT SAFETY

Conflicts: Real-time determinism directly conflicts with multi-tenant throughput. General-purpose schedulers (such as the Linux CFS) maximize overall job throughput and fairness, introducing scheduling jitter of 20–50ms that violates the 10ms hard deadline. Similarly, aggressive battery management throttles CPU frequencies, impeding interactive response times.

Safety-Critical Hierarchy: Bedside monitors are *directly life-critical* (a missed deadline causes physical harm). The central database is *indirectly life-critical* (service interruption halts critical medication verification).

3. HISTORICAL OS LESSONS APPLIED

Batch Systems: Queue-driven, non-interactive execution applies directly to the MRI reconstruction server. **Time-Sharing Systems:** Preemptive time-slicing and memory isolation are vital for clinical workstations and database concurrency. **Mobile Systems:** Dynamic power planes and aggressive process sandboxing are essential for nurses' tablets.

4. ALTERNATIVE CONCEPT & COMPLEXITY ANALYSIS: SINGLE GPOS EVERYWHERE

- **Worst-Case Alarm Latency:** In a GPOS, non-preemptible spinlocks and deferred interrupt handling result in unbounded scheduling latency: $T_{worst} = O(unbounded)$, frequently exceeding 100ms under I/O load and violating the 10ms limit.
- **Space Overhead:** A desktop GPOS requires 4–8 GB RAM and 20 GB storage per node. Across 500 monitors, this wastes over 2 TB of memory and demands active fan cooling, which is unsuited to sterile rooms.
- **Administrative Burden:** Automated rolling updates with mandatory reboots cause unexpected service disruptions on both monitors and the central database.